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Enhancing greenhouse management with interpretable AI: A natural language interface for advanced and optimization-based control

Ramesh Arvind Naagarajan [a,*], Stefan Streif [a,b]

[a] Chair of Automatic Control and System Dynamics, Chemnitz University of Technology, Chemnitz, Germany
[b] Department of Bioresources, Fraunhofer Institute for Molecular Biology and Applied Ecology, Giessen, Germany

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ABSTRACT

As climate change and resource scarcity threaten global food security, greenhouse systems are becoming critical for sustainable agriculture. Advanced control, such as Model Predictive Control (MPC), effectively regulates temperature, humidity, and CO_2 to enhance crop growth and resource efficiency. However, the widespread adoption of such advanced control systems is limited by their lack of interpretability, as growers struggle to understand complex control decisions, particularly during rapid environmental changes. In this work, a Natural Language Generation (NLG) interface has been developed to bridge this gap and transform MPC control decisions into clear, actionable explanations for greenhouse growers. This interface integrates Large Language Models (LLMs) with greenhouse control systems and mathematical constraints, providing a step toward making AI-driven agriculture more accessible. This integration addresses the need for interpretable AI systems in modern agricultural applications. The system allows growers to interact with the control system, query decisions, and receive contextually relevant explanations through Retrieval Augmented Generation (RAG) mechanisms and instruction prompting techniques. The Adaptive RAG (ARAG) framework was evaluated using semantic similarity, information retrieval, and contextual relevance metrics, demonstrating a 12.1% improvement in BERTScore, over baseline methods. These results highlight the system's ability to deliver accurate, well-structured explanations without compromising control performance. By improving the interpretability and accessibility of AI-powered greenhouse automation, this research advances the development of sustainable greenhouse practices that can adapt to the challenges of climate change and resource scarcity. The proposed system represents a significant step toward transforming traditional greenhouse control into more interpretable solutions for modern agriculture.

1. Introduction

In order to meet the demands of a rapidly growing population and the challenges posed by climate change, it is estimated that global food production must increase by 70% by the year 2050 [1]. As controlled environments, greenhouses offer a promising solution to these challenges by enabling year-round crop production and optimizing resource use. However, their complexity increases significantly as greenhouse systems move from simple structures for season extension to advanced, sensor-equipped facilities capable of automatically monitoring and adjusting growing conditions.

Modern greenhouse environments present unique management challenges that extend beyond traditional agricultural practices. Greenhouse operators must simultaneously balance multiple environmental parameters including temperature, humidity, and CO_2 levels, while also managing energy consumption and resource efficiency. These parameters interact in complex ways, creating a multidimensional optimization problem that becomes increasingly difficult to manage using conventional methods [2,3].

In order to address these complex challenges, we introduce the Adaptive Retrieval Augment Generation (ARAG) framework essentially an intelligent translation system that converts complex greenhouse control decisions into practical advice. This approach transforms greenhouse

Abbreviations: AI, Artificial Intelligence; API, Application Programming Interface; ARAG, Adaptive Retrieval Augment Generation; DNN, Deep Neural Networks; EVT, Extreme Value Theorem; GPT, Generative Pre-Trained Transformers; LLM, Large Language Models; ML, Machine Learning; NLG, Natural Language Generation; NLP, Natural Language Processing; NMPC, Nonlinear Model Predictive Control; RAG, Retrieval Augment Generation.

* Corresponding author.
E-mail addresses: ramesh.naagarajan@etit.tu-chemnitz.de (R.A. Naagarajan), stefan.streif@etit.tu-chemnitz.de (S. Streif).

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management by making advanced mathematical solutions understandable and usable for everyday growers. Unlike previous solutions that focused narrowly on generating optimal climate trajectories [4] or providing basic textual representations [5], our framework creates a bridge between sophisticated greenhouse control technologies more accessible and actionable.

The ARAG framework works through two main components: (1) a smart information retrieval system that connects technical control actions to their real-world effects on plants, and (2) an adaptive explanation system that tailors information to match the user's expertise level whether they're a beginning grower or an experienced professional. By finding relevant context and generating explanations that match the user's background, ARAG helps growers understand and trust the advanced technology managing their greenhouse.

Among advanced control methods, Model Predictive Control (MPC) stands out as particularly valuable for greenhouses. Unlike traditional systems that simply react to current conditions (like turning on heaters when it gets cold), MPC works more like a chess player, thinking several moves ahead. It continuously forecasts future greenhouse conditions and plans the optimal sequence of actions to achieve the best growing environment while minimizing resource use. This forward thinking approach allows for more precise management of temperature, humidity, and other factors while balancing various operational goals and practical limitations [4,6,7].

However, a fundamental challenge exists in bridging the gap between the mathematical optimization capabilities of MPC systems and the practical knowledge needs of greenhouse growers. While MPC can compute mathematically optimal control strategies, these decisions often appear counterintuitive to growers who rely on experiential knowledge and traditional practices. For instance, an MPC system might temporarily increase ventilation during cold weather to manage humidity levels a decision that might seem wasteful or harmful without a clear explanation of how this brief energy expense prevents longer-term plant health issues and ultimately saves resources [8,9].

The context-aware retrieval system employs embedding techniques to capture semantic relationships between control parameters and practical growing conditions. It maintains a knowledge base mapping control actions to their practical implications, allowing it to provide explanations that connect mathematical parameters to real-world growing outcomes. The instructional prompting mechanism ensures technical information is presented in an accessible format a novice grower might receive explanations focused on basic cause-effect relationships, while an experienced grower would receive more detailed information about the optimization trade-offs being made. The adaptivity techniques we employed are similar to those described in recent work on instruction-based prompting frameworks [10,11].

By leveraging advanced Natural Language Processing (NLP) and Machine Learning (ML) techniques, this integrated approach translates complex MPC optimization algorithms into intuitive, actionable insights that greenhouse growers can apply to daily cultivation decisions [2]. The objectives of our research can be summarized as follows: (1) to develop a system that makes advanced greenhouse control decisions understandable to growers with varying levels of technical expertise; (2) to bridge the gap between mathematical optimization and practical growing knowledge; (3) to generate adaptive explanations that adjust to user expertise levels; and (4) to enhance trust and adoption of efficient greenhouse control technologies. By achieving these objectives, this research aims to make greenhouse technologies more accessible and practical for real-world agricultural applications, ultimately contributing to more sustainable and efficient food production systems. The contributions of this paper can be summarized as follows:

- A comprehensive ARAG framework that extends beyond reference trajectories translation by integrating contextual aware retrieval and adaptation explanation techniques, enabling real-time interpretation of MPC decisions in response to environmental changes.
- An adaptive explanation component that adjusts the technical complexity of explanations based on user expertise levels, making control decisions accessible to novice and experienced greenhouse growers.
- A semantic retrieval system that connects mathematical control parameters with their practical implications for crop growth, helping users understand how control decisions affect growing outcomes.
- A complete end-to-end system that integrates these components into a unified interface, where previous research that focused only on generating MPC trajectories [4] or basic textual representations [5].

These advancements make sophisticated greenhouse control technologies more accessible and actionable for agricultural growers.

2. Methods

The present study investigates the use of MPC as a predominant advanced control methodology for greenhouse systems, focusing on its capability to predict and optimize greenhouse behavior within a finite time horizon. MPC leverages dynamic models to forecast future greenhouse conditions, allowing the controller to compute optimal reference trajectories for critical environmental variables such as temperature, humidity, and CO_2. This approach is particularly beneficial for balancing crop requirements with resource efficiency. However, despite producing mathematically optimal solutions, these control strategies often remain difficult for greenhouse growers to interpret, highlighting a significant gap between advanced control technologies and their practical implementation.

In order to address this gap, we propose the ARAG system. Fig. 1 illustrates the overall system architecture, while Fig. 2 presents a detailed view of the data processing pipeline derived from the architecture. Together, they provide a comprehensive overview of the system's components and their interactions.

As shown in Fig. 1, when a user query is received, the system initiates a structured data processing flow, as described in Section 2.1. The collected data is first structured and segmented into smaller chunks. These chunks are then processed by an embedding model (green box), which transforms them into vector representations, enabling efficient similarity-based document retrieval. The retrieval module (dark blue box, detailed in Section 2.4) employs context-aware and adaptive ranking strategies (explained in Section 2.4.1) to identify and return the top three most relevant chunks. These selected chunks are passed to the response generator (pink box, described in Section 2.3), which formulates a contextually appropriate explanation. This end-to-end pipeline, encompassing data flow and component interactions, is fully captured in Fig. 1 and expanded upon in Fig. 2.

Following response generation, the system engages the user interaction and feedback mechanism (yellow box, detailed in Section 2.5). If the user approves the generated response, it is finalized and relayed to the chatbot for delivery. However, if the feedback is negative, the system activates a feedback loop (orange box, discussed in Section 2.4.2). In this loop, the retrieved chunks are re-ranked based on the feedback, and the response generation process is iteratively refined to improve accuracy and relevance. This adaptive feedback mechanism, illustrated in Fig. 1, ensures continual improvement and user alignment.

Through this approach, the ARAG system enhances both the transparency and usability of advanced control decisions, thereby bridging the gap between MPC-based algorithms and practical decision-making in greenhouse operations.

2.1. Data preprocessing and document planning

The system processes comprehensive greenhouse MPC data focused on tomato crop cultivation while remaining adaptable to various other

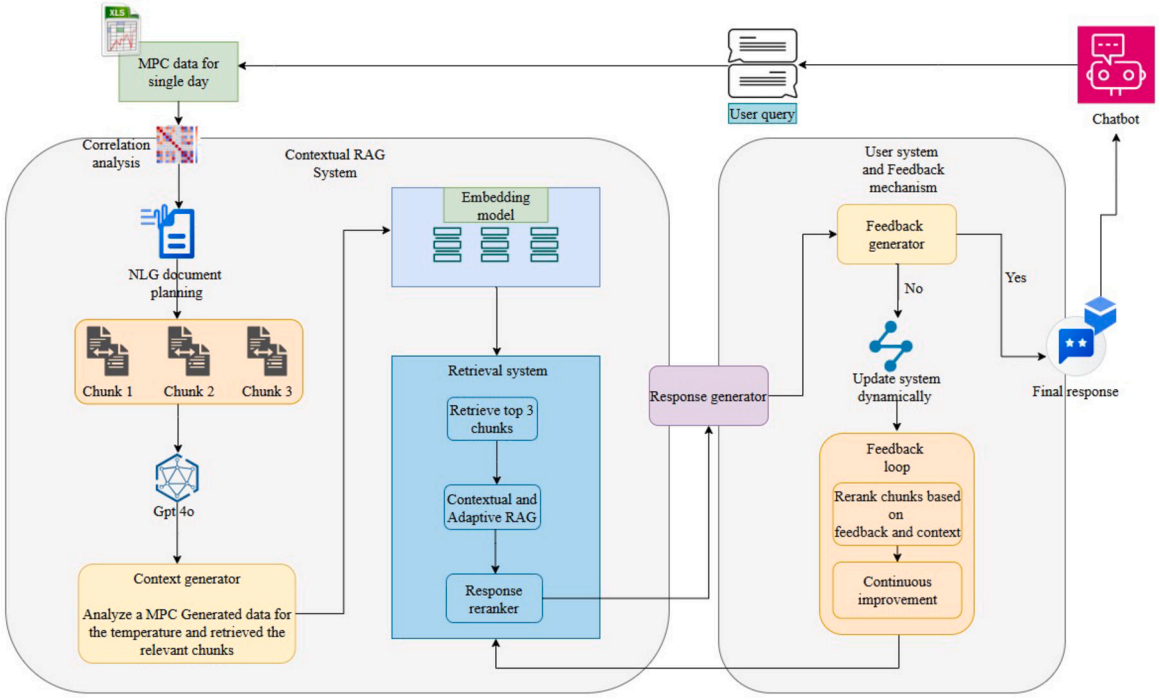


Fig. 1. Architecture of the ARAG System for Context-Aware Chatbots: Comprehensive workflow demonstrating data preprocessing (bottom-left, section 2.1), response generation (pink box detailed in Section 2.3), context-driven retrieval (dark blue box, detailed in Section 2.4), feedback loop (orange box, discussed in Section 2.4.2) and adaptive feedback mechanisms (yellow box, detailed in Section 2.5) for transforming complex MPC data into interpretable explanations.

crops. It integrates internal environmental parameters and external disturbances. The data used in this study originates from the hierarchical control model of Sathyanarayanan et al. [4], where the upper and lower-level optimization problems were solved using real-time weather data collected in 2011 from a greenhouse facility in Berlin. The dataset consists of planned reference trajectories generated at 5-minute intervals, covering a full 24-hour cycle from 00:00 to 24:00. The simulated MPC-generated dataset spans from March to October 2011.

The available variables in the dataset include greenhouse environmental parameters such as temperature, humidity, CO_2 concentration, biomass growth, and operational control actions like heating, cooling, ventilation, and CO_2 injection. Additionally, external disturbances such as outdoor temperature, outdoor humidity, and solar radiation are incorporated to improve the predictive accuracy of the control model.

A key aspect of this system is its ability to analyze and interpret optimal control trajectories while balancing economic objectives and operational constraints. While the dataset is simulated based on MPC decision-making, it does not represent direct real-time greenhouse measurements. However, the system treats these values as dynamically evolving environmental conditions and verifies their relationships with other monitored variables. Whenever a significant change is detected in a primary measurement, the system automatically examines how it correlates with other factors during that period, enhancing the contextual awareness of generated explanations.

In order to enhance and improve trend detection in time-varying greenhouse conditions, a hybrid clustering approach combining K-Means and Extreme Value Theorem (EVT) was initially used in our previous study [5]. However, this method alone may not capture minor but impactful environmental variations. To address this, we introduced a sliding window approach that complements clustering-based trend detection. The system processes a window size of five-time steps (equivalent to 25 minutes at five-minute intervals), applying a threshold of 0.0001 (0.01%) and a gradient step of 0.25 to capture gradual environmental changes.

To further improve trend detection, correlation analysis was employed to assess dependencies among greenhouse climate variables. The system computes Pearson correlation coefficients [12] within a 10-15 minute window to identify relationships between key parameters such as temperature, humidity, and CO_2 levels. Correlations are considered significant when $|r| > 0.7$, ensuring that trends are identified with high reliability. This analysis enables the detection of co-dependent environmental fluctuations and their role in automated control decisions.

The adaptive threshold-based sliding window method also effectively captures slow temperature drifts and minor environmental shifts influencing greenhouse climate control. This approach balances sensitivity and noise filtering, complementing traditional clustering techniques to detect meaningful deviations in environmental parameters over time.

The ARAG framework integrates correlation analysis and gradual change detection techniques to improve contextual accuracy. The NLG document planning component structures the processed greenhouse MPC dataset into well-defined narratives, making complex control strategies more interpretable for growers. Following the approach of Naagarajan et al. [5], this process ensures that raw control data is transformed into structured explanations that clearly outline key environmental transitions and their projected impact. Furthermore, a term-to-feature mapping (as shown in Table 1) is maintained, ensuring precise interpretation of greenhouse control actions. This mapping improves tracking of rapid environmental changes and variable interactions, ensuring generated explanations remain relevant for practical decision-making in greenhouse operations.

2.2. NLG with `GPT-4o`

Once the structured greenhouse MPC data has been processed through document planning, the system requires a robust natural language generation component to convert this structured data into meaningful explanations. In our previous work [5], we utilized `GPT-4`. However, greenhouse control explanations demand both high contextual accuracy and rapid response times to maintain relevance in dynamic growing environments.

In order to address these critical requirements, we transitioned to `GPT-4o` as our primary language model. The decision was driven by the

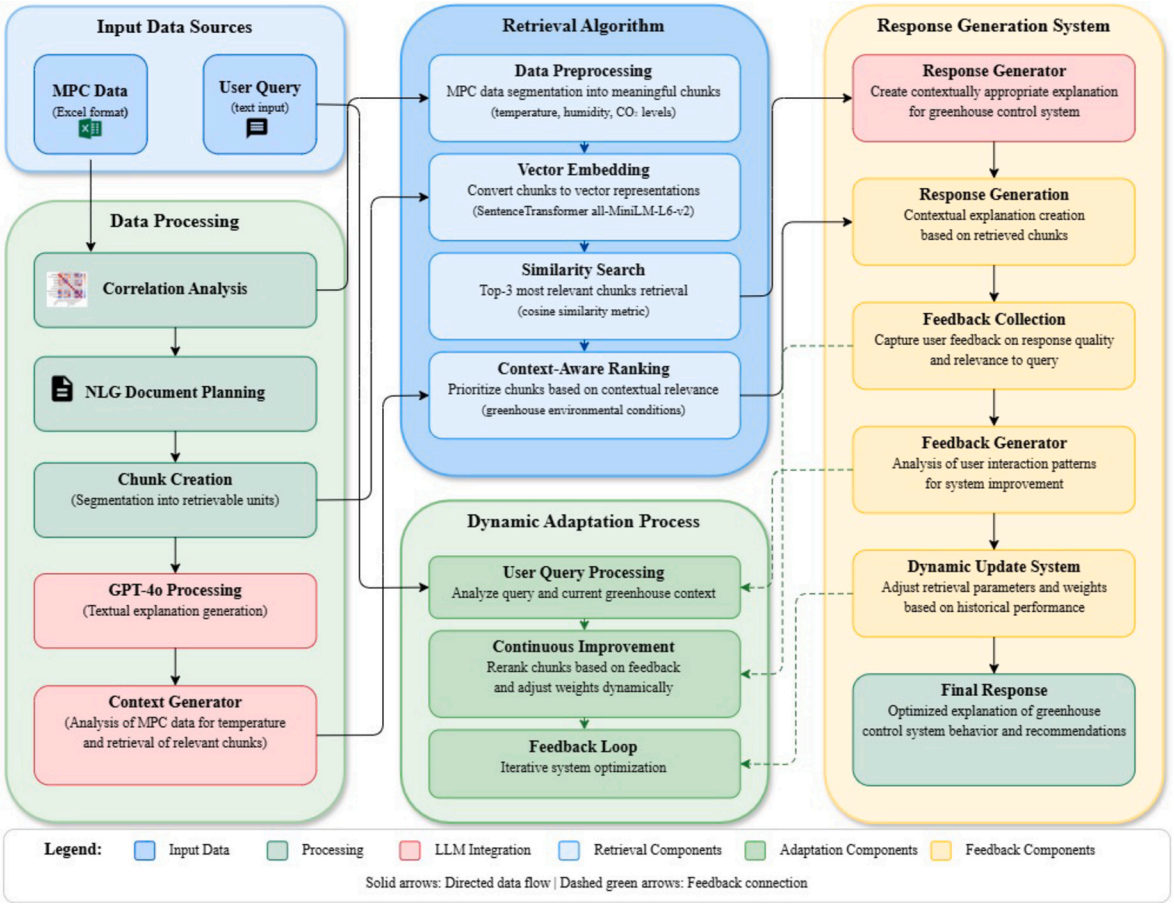


Fig. 2. ARAG System Architecture and Data Processing Pipeline: Detailed breakdown of the system components including Input Data Sources (blue, Section 2.1), Data Processing steps (green, Section 2.1), Retrieval Algorithm components (light blue, Section 2.4), LLM Integration mechanisms (pink, Section 2.3), Dynamic Adaptation Process (green, Section 2.4.2), and Response Generation System (yellow, Section 2.5). The diagram illustrates both direct data flows (solid arrows) and feedback connections (dashed arrows) that enable continuous system improvement based on user interactions.

Table 1
Term-to-Feature Mapping for Natural Language Processing.

Natural Language Term	System Feature
Temperature/Temp	T_{ref}
Humidity	H_{ref}
CO_2/Carbon Dioxide	CO_{2ref}
Biomass/Plant Growth	B_{ref}
Heating	Q_{heat}
Cooling	Q_{cool}
CO_2 Injection	CO_{2inj}
Ventilation	V_{ref}
Outdoor Temperature	T_{out}
Outdoor Humidity	H_{out}
Radiation	R_{sol}
Outdoor CO_2 Level	CO_{out}

Table 2
Performance Comparison: GPT-4o vs. GPT-4 for Greenhouse Control Explanations.

Performance Metric	GPT-4	GPT-4o
Response Time (seconds)	3.82	2.17
ROUGE-L Score	0.73	0.84
BERTScore	0.87	0.92

need for more efficient and contextually aware explanation generation. Table 2 presents a comprehensive performance comparison between `GPT-4` and `GPT-4o` across key metrics relevant to greenhouse control explanations.

The empirical results demonstrate that `GPT-4o` delivers improved performance in explanation generation. Response time decreased by 43.2%, addressing the critical need for timely insights in dynamic growing environments. Simultaneously, explanation quality metrics showed notable improvements: ROUGE-L scores increased by 15.2% and BERTScore improved by 5.4%, indicating more coherent and contextually appropriate explanations.

In order to optimize the model's performance for our greenhouse control domain, `GPT-4o` was configured with a temperature setting of 0.3. This configuration strikes an effective balance between output consistency and adaptive response generation, ensuring that explanations remain factually accurate while flexibly adapting to varied greenhouse conditions. This parameter setting was determined through iterative testing to achieve an optimal trade-off between stability and responsiveness [13,14].

Following the explanation generation, the system embeds the textual output into a vector space using a pre-trained embedding model. This embedding process ensures efficient retrieval of relevant insights when growers query the system about past or current control strategies. The semantic similarity searches enabled by these embeddings ensure that greenhouse operators receive contextually relevant responses when exploring historical control actions and environmental changes. However, embedding-based retrieval alone can struggle with time-sensitive greenhouse conditions that require rapid adaptation to changing variables. To address this limitation, the system incorporates a context-aware retrieval mechanism that dynamically adjusts to real-time changes in

greenhouse parameters, further enhancing the practical utility of the generated explanations.

2.3. The general framework for context-aware RAG system architecture

Building upon the structured retrieval process introduced, which uses embeddings for document storage and similarity-based retrieval, this component integrates RAG to accelerate response times and enhance adaptability to real-time greenhouse fluctuations.

RAG is a powerful way to improve LLMs by adding external information while keeping responses accurate and based on facts [15]. Traditional RAG systems combine search tools with AI language models to produce better and more relevant responses. However, in the context of greenhouse control systems, conventional RAG methodologies struggle to handle the constantly changing nature of environmental variables, control input, and their impacts on crop growth [16].

The ARAG framework extended the traditional RAG architecture to address these challenges within the agricultural domain. By incorporating dynamic context adaptation mechanisms [17] and real-time feedback integration [18], the system has evolved its understanding of greenhouse control decisions continuously. This adaptation is particularly crucial for interpreting MPC decisions, where control actions had to be explained in terms of their immediate and long-term impacts on crop growth [19].

2.4. Context-driven information retrieval in RAG

This section discusses how greenhouse control data is retrieved and processed to generate meaningful explanations using the ARAG framework is detailed in this section. Context-driven information retrieval plays a crucial role in extracting the most relevant knowledge from the available greenhouse dataset, ensuring that responses align with the grower's queries.

Within the ARAG framework, the context-aware retrieval model is pivotal in leveraging advanced data processing and embedding techniques to manage greenhouse control data. The system is designed to understand and explain MPC decisions by showing how different control actions aim to increase production, adapt to real-time weather conditions, and minimize operational costs.

2.4.1. Chunk-based retrieval algorithm

Once the raw data is collected and structured, a chunk-based retrieval algorithm facilitates the organization and retrieval of MPC control data, ensuring the precise retrieval of relevant information. This approach transforms raw control trajectories into searchable knowledge units [20]. These documents represent a look-ahead control approach, where the controller plans future actions based on its predictive model.

In order to handle expected changes in the environment, the system incorporates forecasted external weather conditions from meteorological services as input parameters to the MPC model. This allows the controller to adapt its decisions over a 24-hour prediction horizon. These predicted trajectories containing planned control actions and their expected impact on the greenhouse environment are then divided into fixed-size chunks using a sliding window approach. Each chunk represents a segment of time-series data for specific features, such as temperature, humidity, and other environmental states. This approach ensures the preservation of temporal coherence and the maintenance of semantic relationships between various control parameters [21,22].

Each chunk is embedded into dense vector representations using the SentenceTransformer model, specifically the `all-MiniLM-L6-v2` variant [23]. This model is selected for its optimal balance between computational efficiency and semantic representation capability, making it suitable for the ARAG framework. The embedding is performed using the same model upon receiving a user query. The system computes cosine similarity between the query embedding and the stored chunk embeddings to identify the most relevant information [24,25]. This similarity-based retrieval method ensures that retrieved documents align with the greenhouse grower's specific queries.

2.4.2. Dynamic adaptation of retrieval strategies

The retrieval mechanism serves as the foundation for identifying relevant greenhouse control data. Furthermore, to ensure more effective query responses, the system must adapt dynamically to changing greenhouse conditions and user preferences. The output of the chunk-based retrieval algorithm is used as input for dynamic adaptation, ensuring that the retrieved documents are continually refined based on real-time feedback.

As illustrated in Fig. 2, the retrieval algorithm and dynamic adaptation processes work together to deliver contextually relevant information. The retrieval algorithm processes greenhouse MPC data through data preprocessing, vector embedding, similarity search, and context-aware ranking, ensuring only the most relevant information is selected for user queries.

The dynamic adaptation process enables continuous improvement through user feedback. This process analyzes query content and current greenhouse context, generates explanations based on retrieved information, and captures user satisfaction with provided explanations. This feedback informs the continuous improvement process that reranks chunks based on user feedback and contextual factors.

Key adaptation mechanisms The system incorporates two main adaptation mechanisms:

1. **Feedback-Driven Weights:** As shown in the "Continuous Improvement" component (bottom right of Fig. 1), user feedback is used to adjust the weights of chunks. This improves retrieval accuracy over time. For example:
   - **Positive Feedback:** If a greenhouse operator provides positive feedback on an explanation about temperature control during high radiation periods, the system increases the relevance weight of chunks containing similar environmental patterns by 0.15 points on a scale of 0 to 1.0.
   - **Negative Feedback:** Negative feedback leads to a decrease in chunk relevance by 0.1 points, gradually steering the system away from less helpful information patterns.
2. **Contextual Awareness:** The *User Query Processing* and *Response Generator* components in Fig. 1 implement contextual awareness, adapting explanations based on the current greenhouse conditions and constraints (e.g., soft or hard constraints) [26].

By integrating these adaptation mechanisms, the ARAG framework enhances retrieval precision and ensures that responses evolve in tandem with dynamic greenhouse conditions.

2.5. Instruction prompting strategies for RAG optimization

An effective retrieval algorithm is essential, but the true value of the proposed system lies in its ability to generate explanations that greenhouse growers can readily understand and apply. The instruction-prompting system has significantly enhanced the ability to make interpretable control decisions for greenhouse growers [27].

Building on recent developments in prompt engineering [28], a multilevel prompting strategy was adapted to accommodate complexity based on user expertise and specific query contexts. The system employs a hierarchical reasoning framework [29] to break down MPC decisions into comprehensible steps. Each explanation is structured to provide both immediate operational insights and long-term strategic understanding.

Role-based prompting represents a key enhancement, enabling adaptive responses tailored to different greenhouse management roles. This approach customizes the chatbot's response depending on the user's role:

1. **Operator Level**: For day-to-day operational staff, explanations focus on immediate actions and practical implications. For example, when explaining temperature adjustments, the system might respond: *"The temperature is being increased to 24 °C over the next 30 minutes to promote fruit development while minimizing energy usage during the forecasted cloudy period."*
2. **Supervisor Level**: For management personnel, explanations include operational rationales and economic considerations: *"The temperature increase to 24 °C optimizes the growth-to-energy-consumption ratio, providing an estimated 3% increase in fruit development while maintaining energy usage within daily budget constraints. This decision factors in the weather forecast indicating reduced solar radiation for the next 4 hours."*

These role-based prompts are implemented through template structures containing specific parameters for detail level, technical terminology, and explanation scope. The prompting mechanism also incorporates agricultural domain knowledge through carefully curated templates and rules that ensure technical explanations remain grounded in practical farming contexts while maintaining scientific accuracy [30].

2.6. Evaluation methodology for ARAG performance

The proposed ARAG system's performance evaluation employed several techniques to assess retrieval efficiency and response quality, with a specific focus on greenhouse control applications and tomato cultivation outcomes. The evaluation framework includes a baseline system, retrieval evaluation, response quality assessment, and operational impact assessment.

The implementation of data analysis procedures utilized Python as the primary programming language, leveraging specialized libraries such as `pandas` and `numpy` for efficient data manipulation and statistical computations. The visualizations were created using `matplotlib` for better data representation, and responses were retrieved using the `GPT-4o` API for NLP tasks. The chatbot interface was developed using `Streamlit` to provide an interactive user experience.

A benchmark for performance measurement was required to evaluate the proposed approach effectively. The baseline system implemented a simplified version of the retrieval and explanation generation process, designed to highlight the limitations of traditional approaches. For data preprocessing, it used keyword-based chunking, where the greenhouse dataset was divided into fixed-size chunks based on predefined keywords (e.g., "temperature," "humidity," "CO_2"). Unlike the ARAG framework, it did not generate embeddings or consider semantic relationships between chunks.

The baseline retrieval mechanism relied on exact keyword matching. For instance, if a user queries, *"Why is the temperature set to 25 °C?"*, the system retrieves chunks containing the keyword 'temperature,' regardless of contextual relevance or semantic similarity. Explanations were generated using fixed templates, such as *"The [feature] is set to [value] for optimal growth."* In contrast, the ARAG framework employs an LLM to generate dynamic, context-aware responses.

While the baseline system provided a structured comparison, it exhibited significant limitations. Its fixed keyword-based retrieval and static template responses resulted in generic, contextually rigid outputs that failed to adapt dynamically to user queries. Moreover, it lacked semantic understanding and the ability to incorporate real-time feedback, making it insufficient for greenhouse management scenarios requiring more adaptive decision-making.

These identified shortcomings directly informed the design of the ARAG framework. To address them, ARAG integrates semantic search mechanisms, dynamic response generation, and contextual adaptation powered by an LLM. The improvements ARAG introduced were measured using quantitative metrics (retrieval accuracy, ROUGE-L, BERTScore) and qualitative user feedback, confirming notable enhancements in response clarity, contextual relevance, and usability [31,32].

2.6.1. Metrics for retrieval effectiveness

The retrieval mechanism's effectiveness was evaluated through a comprehensive approach combining multiple quantitative metrics. The proposed system employed Top-3 retrieval accuracy assessment, implementing cosine similarity matching between query and stored embeddings [26,33]. This approach ensured that the most relevant greenhouse control data was consistently retrieved to generate accurate MPC explanations.

A strict similarity threshold of 0.85 was maintained to ensure high-quality context selection, guaranteeing that only the most relevant explanations were retrieved. This threshold was selected based on prior studies demonstrating its effectiveness in balancing the inclusion of pertinent data while filtering out irrelevant content. This parameter also played a role in assessing Top-3 retrieval accuracy, ensuring that retrieved information maintained high precision. For instance, researchers have employed this threshold in clinical information extraction to maintain high-quality context selection, ensuring that retrieved explanations meet established quality standards [34].

Similarly, in evaluating RAG systems, this threshold has been used to validate matched content from the knowledge base, enhancing the system's reliability. By adopting this approach, our system ensures that only the most relevant information contributes to generating MPC explanations, preventing the inclusion of low-relevance content.

The dynamic document planning process built upon these correlations, implementing a refined structuring mechanism considering feature importance and temporal relevance [35]. This approach ensured that generated explanations maintained logical coherence while prioritizing the most influential environmental factors. The planning process adapted to changing greenhouse conditions and seasonal variations, ensuring that explanations remained relevant and practical throughout the growing cycle [28].

2.6.2. Measuring operational impact of RAG systems

The operational impact of the system was measured through its influence on greenhouse management decisions and outcomes. As shown in Table 1, the feature-based mapping was employed to understand how environmental parameters affected explanation quality and decision-making effectiveness.

User feedback was integrated as an essential component of the evaluation framework, which was developed to track immediate response effectiveness [14] and long-term improvement in decision quality. We implemented a structured feedback collection system to quantify this impact using a simple yes/no/partial response mechanism to measure explanation clarity, relevance, and actionability. This ternary feedback approach simplified the interaction while still providing actionable data. Based on every 50 responses, the system adjusted the weights of the candidate chunks and fine-tuned the responses to improve their accuracy and relevance. This iterative feedback loop demonstrated assistance in the refinement of the system over time, ensuring that the explanations evolved to meet the users' needs more effectively.

The feedback loop enabled continuous refinement of the explanation generation process, ensuring the system evolved to meet the practical needs of greenhouse growers. Every negative feedback instance triggered a re-ranking of candidate chunks. It adjusted the weighting factors by a factor of 0.85 (reducing previously selected chunks' relevance), while each positive feedback increased weights by a factor of 1.15 for similar environmental patterns.

By integrating these evaluation components, the system was ensured to retrieve relevant contextual data, generate accurate explanations, and contribute meaningfully to improved greenhouse operations and crop management decisions. The evaluation approach employed in this system constituted the integration of quantitative metrics, providing a foundation for validating the system's effectiveness in controlled greenhouse environments.

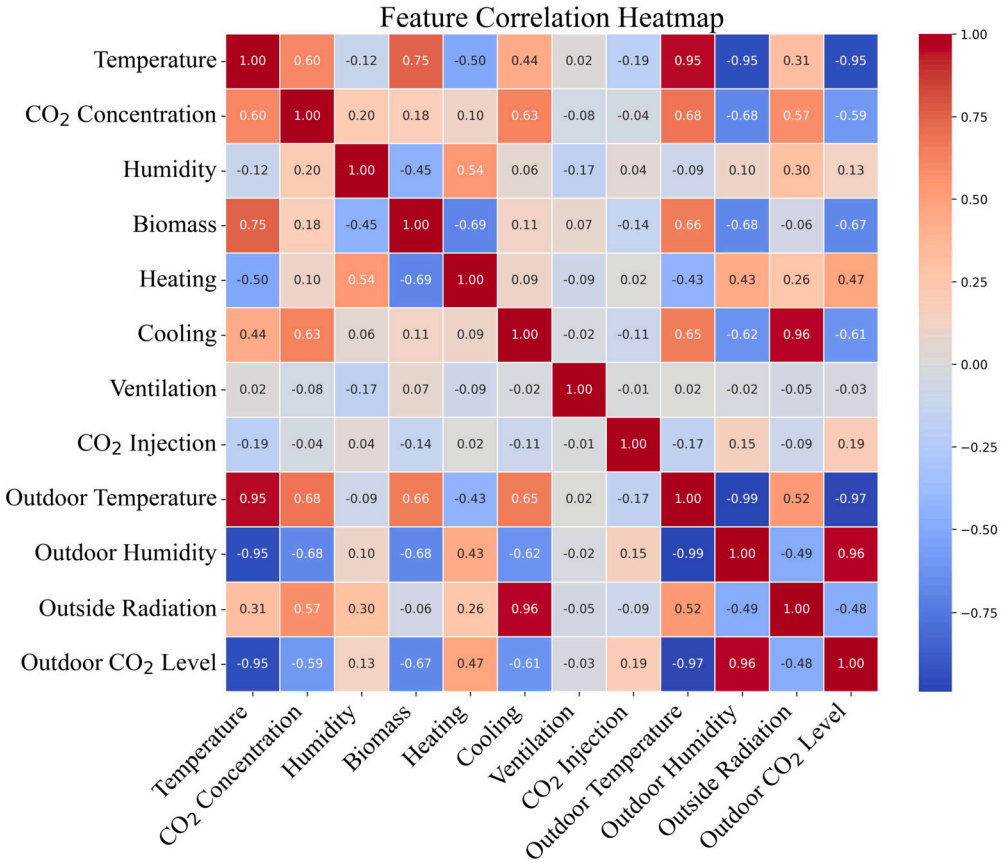


Fig. 3. The Correlations of features in Greenhouse Climate Control.

3. Results

The results are structured as follows. First, we describe the enhanced document planning and data analysis system. Next, we present comparative model evaluations, improved retrieval mechanisms, multistep prompting frameworks, and detailed performance metrics. The system generates contextually rich and accurate environmental analyses by integrating correlation analysis, gradual change detection, and multilayered prompting strategies.

3.1. Document planning and data analysis system

The enhancement of the document planning and data analysis model involved integrating NLG-driven strategies into the greenhouse climate control framework, expanding upon the approach introduced by Naagarajan et al. [5]. This framework, referred to as the Adaptive Greenhouse Control System, leverages NLG to improve decision-making and explainability in greenhouse automation.

Implementation of two complementary approaches enabled comprehensive analysis of environmental changes: correlation analysis to identify strong relationships between variables and gradual change detection to capture subtle variations over time. The reason for employing both techniques is that strong correlations help in identifying key control dependencies, while subtle variations assist in understanding evolving climate conditions that influence control decisions. These insights enable the generation of textual explanations that justify specific control actions taken by the system, thereby improving transparency and operator trust.

3.1.1. Correlation analysis for trend change

The correlation analysis examined significant dependencies among climate control variables, particularly the strong relationships between temperature, humidity, and CO_2 levels. These correlations directly influenced the automated control actions, such as adjusting ventilation and irrigation. Meanwhile, the gradual change detection method identified nuanced shifts in environmental variables that were not evident in correlation analysis alone. This method was particularly useful in detecting slow temperature drifts that might require long-term adjustments rather than immediate control actions.

Fig. 3 illustrates the impact of correlation analysis and gradual change detection on greenhouse operations. The panel visualizes strong correlations between key variables, indicating how different factors influence greenhouse conditions shown in the below Section 4.4. By integrating these analyses, the system provides a more comprehensive understanding of greenhouse dynamics.

3.1.2. Gradual change detection for subtle variations

The adaptive threshold-based sliding window method effectively detected minor environmental variations over time. Unlike traditional clustering methods, which may overlook gradual shifts, this approach captured slow environmental transitions critical for long-term greenhouse optimization. These gradual changes included slow drifts in greenhouse temperature or CO_2 concentration that, while not triggering immediate corrective actions, influenced control strategies over extended periods.

By detecting subtle environmental changes, the system ensures a deeper understanding of greenhouse conditions, providing an important foundation for evaluating its performance against the baseline model.

3.2. Baseline model for evaluation

A baseline system was established for proper assessment of the enhancements, implementing a keyword-based retrieval and template-driven response mechanism utilizing the `all-MiniLM-L6-v2` embed-

ding model. While this baseline approach offered computational efficiency, it encountered significant limitations in contextual adaptation and dynamic trend analysis. Despite its processing speed, the baseline system struggled to analyze shifting trends and adapt to context effectively. The limitations of the rigid framework were evident in its inability to identify minor patterns, and the system was incapable of detecting relationships between variables or assessing environmental changes in real-time.

3.3. GPT-4o model

The comparative analysis demonstrated that GPT-4o outperformed its predecessor in several key areas, including generating precise, structured responses aligned with greenhouse monitoring requirements, the reduction of inference time, and the improvement of real-time document generation capabilities [14]. The model's enhanced contextual retention enabled comprehensive multi-feature trend analysis while maintaining reproducibility and scalability across domains requiring structured, data-driven summaries. Building upon the enhanced model's contextual understanding, we enhanced the document retrieval process using advanced chunking strategies and optimized embeddings.

3.4. RAG and embeddings for document retrieval

In order to address the limitations of the baseline retrieval system, the RAG implementation enhanced document retrieval through an optimized chunking strategy. A window size of 200 tokens with a 50-token overlap was employed. In this context, tokens refer to smaller units of text (such as words or word pieces) that enable the model to process and retrieve relevant information efficiently. These parameters were determined through empirical analysis to maximize retrieval accuracy while minimizing information loss.

The retrieval process leveraged feature-specific embeddings generated using the all-MiniLM-L6-v2 model and incorporated weighted cosine similarity scores for multi-step retrieval. Cosine similarity, a mathematical measure that calculates the angle between two vectors in multi-dimensional space (where smaller angles indicate greater similarity), allowed the system to identify how closely related a user's question was to stored greenhouse information [36]. This approach allowed the system to retrieve the top three relevant chunks per requested feature, maintaining strong contextual coherence while ensuring greenhouse trend analysis remained accurate and interpretable.

A comprehensive assessment methodology was developed to evaluate the quality of the generated explanations, combining automated metrics with expert validation. The automated assessment utilized ROUGE-L scores to test the lexical overlap between generated responses and expert-annotated reference explanations [31]. This was supplemented with BERTScore analysis to extract semantic similarity beyond exact word matches by leveraging contextual embeddings [32]. These automated metrics provided quantifiable information on the linguistic quality and semantic accuracy of the produced explanations.

The expert validation process was conducted by research domain specialists who assessed the explanations based on technical accuracy, practical relevance, and clarity. This validation was based on feedback obtained through chatbot interactions, where domain experts reviewed generated responses for alignment with greenhouse management best practices. The validation involved a sample of 100 expert evaluations to ensure a robust assessment of the system's performance.

3.5. Top-K retrieval and feedback adaptation for relevant chunks

The system presented the three most relevant chunks of information by looking at how well the features match and how closely they relate to the question. The system adjusted the scoring to account for length to ensure longer sections do not unfairly dominate the results.

Feature Analysis

Features:
Analyze: {feature_list}
Timeframe: {time_period}

Analytical Parameters:
Initial State: {starting_conditions}
Final State: {ending_conditions}
Threshold Values: {min_max_parameters}
Pattern Recognition: {trend_pattern}

Output Requirements:
Unit System: {unit_system}
Numerical Precision: {decimal_places}
Anomaly Flagging: Values beyond {threshold_values}

Fig. 4. Base prompting template structure for environmental feature analysis showing the key components and their relationships.

When combining these scores to determine which chunks were most useful, the system also considered past user feedback to ensure a balanced representation across different features [26], creating a more responsive and adaptive retrieval mechanism. To measure the success of these retrieval mechanisms, the following section presents the quantitative metrics used to assess the effectiveness of the retrieval and feedback adaptation process. With the optimized chunking and embedding strategies, the next step was to evaluate the system's prompting techniques.

3.6. Multi step prompting techniques

Building on the retrieval mechanisms described above, we implemented a multi-step prompting approach to enhance the system's response quality further. The ARAG model utilized a structured, multi-level prompting approach customized for various research fields to enhance further the system's capability to generate contextually appropriate and accurate information. Beginning with fundamental analytical methods, more specialized domain knowledge was gradually integrated. This combination of basic prompts with targeted information retrieval ensured consistent and accurate results across different areas of study.

3.6.1. Base prompting

The base prompting layer was the fundamental framework for analytical operations, establishing core principles that ensured consistency and structural integrity across all analyses. A standardized template structure was implemented to maintain unit consistency while enabling comprehensive trend reporting, as illustrated in Fig. 4.

The base prompting framework consisted of several structured sections. The Features section outlined the core components to be analyzed including the feature_list of greenhouse parameters (e.g., temperature, humidity) that influence decision-making. Each feature contributes to the overall understanding of environmental conditions, while the time_period set the time frame for the analysis (e.g., hourly, daily).

The Analytical Parameters section included the baseline and final measurements, which were captured through starting_conditions and ending_conditions. Additionally, it included the acceptable range of values for the parameter through min_max_parameters and the expected trend behavior over time with trend_pattern (e.g., increasing, decreasing).

The Output Requirements section ensured that the results were presented precisely. The unit_system defined the measurement units (e.g., metric or imperial), while the decimal_places specified the precision for numerical results. The threshold _values were essential for identifying anomalies that fell outside the expected range, flagging

Query Processing: Template for Query Processing Analysis

Input: {user_question}
Context: {domain_context}
Constraints: {domain_rules}

Retrieval Parameters:
Chunk Size: {chunk_parameters}
Relevance Threshold: {similarity_score}
Feature Focus: {target_features}

Response Structure:
Observation: {data_pattern}
Analysis: {causal_factors}
Recommendation: {action_items}

Context Integration:
System Context: {system_context}
Historical Data: {relevant_history}
Environmental Factors: {external_conditions}

Fig. 5. Advanced prompting template for query processing and response generation.

values that deviated from the norms. The base prompting layer served as the foundation for structured analysis, ensuring consistency and reliability across all retrieved data, and it set the stage for more advanced retrieval-augmented generation.

3.6.2. Advanced RAG prompting

Building on the base prompting framework, the advanced RAG architecture introduced additional layers of contextual processing and adaptive response mechanisms to enhance accuracy and domain specificity. As illustrated in Fig. 5, this enhancement allowed for more precise information extraction while maintaining structural integrity, making the system more effective at handling complex queries.

The Query Processing section component analyzed the user's initial question and relevant domain context, ensuring a comprehensive interpretation of the inquiry. The `user_question` variable captured the exact query or input from the user. The `domain_context` referred to the background knowledge or area of interest in which the query was situated, ensuring that the response aligned with relevant domain expertise. The `domain_rules` defined the constraints or limitations for the query processing.

The Retrieval Parameters section established critical processing guidelines for efficient data segmentation and filtering. The `chunk_parameters` dictated how the input was divided into smaller pieces for more efficient analysis. The `similarity_score` represented the threshold for how relevant a particular chunk of data needed to be about the user's query. The `target_features` highlighted specific aspects of the query or dataset that were of primary importance.

The Response Structure section provided the format for presenting the results. The `data_pattern` referred to trends, patterns, or important observations that emerged from the data. The `causal_factors` described the underlying causes or correlations that explained the identified data patterns. The `action_items` suggested specific steps, decisions, or interventions that should be taken based on the analysis.

In the Context Integration section, the `system_context` captured the setup and settings of the greenhouses system used for analysis. The `relevant_history` included historical data or past patterns that provided context to the current analysis. The `external_conditions` accounted for any environmental factors that might influence the analysis.

Table 3
Performance comparison of user responses: Baseline vs ARAG System.

Metric	Baseline System	ARAG System
ROUGE-L	0.05	0.07
BERTScore	0.68	0.79
Relevance	0.41	0.57

3.7. Performance metrics

The performance evaluation was conducted using 100 sample responses collected from users engaged with the system. These responses were assessed using three key metrics impacting greenhouse management effectiveness: ROUGE-L, BERTScore, and Relevance. ROUGE-L evaluates how well the system captures the sequence and structure of information critical for time-sensitive greenhouse operations, where the proper sequence of control actions directly affects crop health. BERTScore measures semantic similarity between generated responses and expert references, ensuring that explanations maintain technical accuracy regarding critical parameters like temperature ranges and humidity levels that determine optimal growing conditions. The Relevance score quantified how well the generated response addresses the specific greenhouse management query, directly impacting a grower's ability to make informed operational decisions. As shown in Table 3, the ARAG system demonstrated significant improvements over the Baseline System in all evaluated metrics.

The ROUGE-L score, which evaluated the longest common subsequence between the generated response and reference, shows an improvement of 40.38%. This substantial increase translates to more comprehensive explanations of control sequences in greenhouse management, allowing operators better to understand the timing and progression of environmental adjustments. Similarly, BERTScore, which evaluated the semantic similarity between the generated response and reference using contextual embeddings, improved by 12.1%, ensuring that explanations about plant physiology and environmental control systems maintained scientific accuracy. Lastly, the Relevance score, which measured how directly the generated response addresses the specific greenhouse management query, showed an improvement of 38.3%.

In order to establish the statistical significance of these performance improvements, we conducted a Wilcoxon signed-rank test on the evaluation metrics. This non-parametric test was selected because it makes no assumptions about the underlying distribution of the data and is appropriate for comparing paired samples [37,38]. For each of the sample responses, we analyzed the paired differences in ROUGE-L, BERTScore, and Relevance scores between the baseline and ARAG systems. The Wilcoxon signed-rank test revealed statistically significant improvements across all metrics: ROUGE-L ($Z = -5.97$, $p < 0.001$, $r = 0.42$), BERTScore ($Z = -6.33$, $p < 0.001$, $r = 0.45$), and Relevance score ($Z = -6.81$, $p < 0.001$, $r = 0.48$). The effect sizes $r = \frac{Z}{\sqrt{N}}$ indicate medium to large practical significance according to Cohen's guidelines [39], with the strongest effect observed in the Relevance metric.

In addition, the 95% confidence intervals were determined utilising the Hodges-Lehmann estimator [40]: ROUGE-L [0.014, 0.024], BERTScore [0.087, 0.128], and Relevance [0.130, 0.182]. These intervals exclude zero, further confirming that the improvements are statistically significant. The Wilcoxon test is computationally efficient and can be quickly recalculated if needed, making it ideal for iterative system evaluation and validation [41].

These statistical findings, summarized in Table 4, provide evidence that the ARAG system's enhanced performance represents improvements rather than random variation, with particularly strong gains in contextual relevance for greenhouse management applications.

These performance metrics highlight the significant improvements in retrieval and response quality for practical greenhouse management

Table 4
Statistical significance testing results comparing baseline and ARAG systems.

Metric	Z-score	p-value	Effect size (r)
ROUGE-L	-5.97	<0.001	0.42
BERTScore	-6.33	<0.001	0.45
Relevance	-6.81	<0.001	0.48

Table 5
Instruction Prompting Effectiveness: Base vs Refined Prompting.

Metric	Base Prompting	Refined Prompting
ROUGE-L	0.06	0.07
BERTScore	0.76	0.79
Relevance	0.51	0.57

"The temperature is currently at 16 °C, and the trend is increasing in the evening around 5 PM. The system has detected minor daily fluctuations, with temperatures between 20 °C and 24 °C. Based on historical data, we expect the temperature to decrease after sunset. The greenhouse ventilation system operates normally, and no temperature alerts have been triggered. Users should monitor the temperature changes during the peak hours between 2 PM and 6 PM."

Fig. 6. Temperature analysis from the baseline RAG system demonstrating limited contextual awareness and template-based response generation.

applications, which were further enhanced by the multi-layer prompting techniques outlined in the following section.

The effectiveness of the multi-layer prompting strategy was evaluated by comparing the performance of the base and refined prompting methods. As shown in Table 5, the refined prompting strategy significantly improved ROUGE-L, BERTScore, and Relevance over the base prompting.

3.8. The effectiveness of instruction prompting

The refined prompting strategy outperformed the base prompting strategy in all three metrics. The ROUGE-L score improved by 14.3%, BERTScore improved by 4.59%, and Relevance improved by 11.33%. The findings indicated that the multi-layer prompting approach was more effective in generating high-quality responses that exhibited enhanced semantic alignment and relevance to the user's query. These improvements highlight how multi-layer prompting enhances response quality, alignment with the user's query, and overall relevance.

3.9. The comparison of baseline and ARAG model

In order to demonstrate the effectiveness of the ARAG system, a comparison was made between its performance and that of the baseline system using a common user query *"What is the temperature trend today?"*.

As illustrated in Fig. 6, the baseline system, which relied on keyword-based retrieval and template-based responses, yielded the following response to the query. Conversely, the ARAG system (see Fig. 7) generated a more detailed and contextually rich response to the same query.

The qualitative differences between these responses are evident in their information retrieval and contextual awareness. While the baseline provides generic information with limited details, the ARAG system offers precise temperature ranges, specific timing of fluctuations, and operational context regarding environmental constraints. The baseline system relied on simple keyword matching (finding exact words without understanding meaning). In contrast, the ARAG system used semantic understanding (comprehending the intent behind questions) to generate more helpful greenhouse management insights. These improvements are further confirmed by quantitative evaluation, where the ARAG system achieved better performance metrics than the baseline for this specific query, as shown in Fig. 8.

"Throughout the day, the indoor temperature in the Venlo-type greenhouse follows a pattern of fluctuations with several sharp increases and decreases. Starting from an initial value of 16 °C at the beginning of the day, the temperature undergoes multiple cycles of rising and falling, reaching a peak value of 22 °C at 15:30. The lowest temperature recorded today was 14 °C at the start. By the end of the observation period at 23:50, the temperature settles at 16 °C. Generally, the temperature demonstrates a fluctuating pattern with frequent sharp increases and decreases, staying mostly within the MPC soft constraint range of 18 to 26 °C but occasionally approaching the hard constraint boundary of 14 °C in the early hours. This dynamic trend can be attributed to the interaction of environmental control systems, such as heating and cooling, which manage the internal conditions in response to changing external factors."

Fig. 7. Detailed temperature analysis from the ARAG system showing daily temperature fluctuations, constraints, and environmental control system interactions in the Venlo-type greenhouse.

System Performance Metrics

ROUGE-L: 0.07 (Baseline: 0.04)
BERTScore: 0.81 (Baseline: 0.68)
Relevance Score: 0.57 (Baseline: 0.41)
Response Time: 21.1 sec (Baseline: 8.4 sec)

Fig. 8. The Performance measure of the single user query prompting based on the Baseline and the ARAG.

The evaluation demonstrated that the ARAG system significantly improved performance across all key metrics compared to the Baseline System. These results indicate that the ARAG system improves retrieval quality, semantic coherence, and contextual accuracy, although at the cost of increased response time. These single-query performance results provide a more granular look at the system's efficiency, complementing the broader evaluation of retrieval and prompting techniques.

4. Discussion

The following discussion elaborates on three important findings: the proposed system's adeptness in handling complex questions, the depth of its contextual responses, and its adaptability across different research domains.

4.1. Contextual awareness and query understanding

One of the most significant improvements in the ARAG system is its enhanced contextual awareness, allowing it to interpret and respond to user queries accurately. For instance, when a user asks, *"What is the temperature trend today?"* The baseline system often struggles to identify which specific temperature is being referenced, whether it is the outdoor temperature, indoor greenhouse temperature, or another related metric. The limitation results from the baseline system's reliance on predefined templates and basic keyword matching, which cannot distinguish between similar terms or infer context. Because of this, the baseline system can give general or insufficient answers that do not consider the user's purpose.

In contrast, the ARAG system leverages sophisticated embedding models and context-aware retrieval mechanisms to identify the relevant

temperature metric accurately. By analyzing the semantic context of the query and integrating term-to-feature knowledge, the system can distinguish between different types of temperatures and provide precise, detailed responses that directly support agricultural decision-making. This capability is crucial for greenhouse growers who must maintain optimal growing conditions and quickly address environmental fluctuations that could impact crop health and yield.

For instance, in Venlo-type greenhouses, which are widely used in commercial horticulture, effective crop management requires more than just simple temperature readings. While the baseline system may report current temperature values, it lacks the ability to provide insights into how these values fluctuate over time, what causes these variations, and how they interact with environmental control systems. The Adaptive RAG system addresses these gaps by analyzing temperature data in the context of broader agricultural conditions, considering peak measurements, underlying causes, and interactions with climate control mechanisms.

The proposed greenhouse operation-specific system requirement lets growers decide on irrigation scheduling, nutrient application, and climate control adjustments. The system achieves this level of detail through its ability to retrieve and analyze relevant data chunks using a sliding window approach, generate embeddings that capture the semantic context of the data, and incorporate greenhouse-specific domain knowledge into the response generation process. The result is a precise environmental management guide that has the potential to enhance resource efficiency and crop productivity in controlled environment agriculture.

4.2. Baseline prompting vs. advanced prompting

The comparison between baseline prompting and advanced prompting provides valuable insights into the effectiveness of different approaches to agricultural information systems. The baseline system, implemented as an evaluation benchmark, employs a standard approach using predefined templates and keyword matching. This computationally efficient approach provides basic information that meets fundamental user needs.

For instance, when analyzing temperature trends, the baseline system effectively delivered essential information, including the current temperature and a general trend description, as shown in Fig. 6. While this information satisfies basic monitoring requirements, the evaluation revealed opportunities for enhanced detail, such as specific temperature values at different times of the day, explanations of underlying causes, and references to greenhouse models. This evaluation was conducted through sample user experience feedback and expert assessment, ensuring the insights align with real-world agricultural decision-making needs. The baseline system serves its intended purpose as a reference point against which to measure advancements in agricultural information delivery.

The advanced prompting strategy employed by the ARAG system builds upon these fundamentals to deliver enhanced analytical capabilities, as shown in Fig. 7. Using a multi-layer prompting approach that progresses from fundamental analytical principles to sophisticated domain-specific applications, the ARAG system expands on basic information to provide comprehensive insights, including specific temperature values at key times, detailed temperature pattern descriptions, references to the greenhouse model, and explanations of the underlying causes of observed trends.

The enhancement in analytical depth is attained using complementary techniques, including targeted data chunk retrieval, semantic embedding generation, and the incorporation of greenhouse-specific domain knowledge into the response generation process. The evaluation demonstrates the potential of these enhancements to provide greenhouse growers with more comprehensive information for decision-making while acknowledging the baseline system's effectiveness for basic monitoring tasks.

Furthermore, the impact of advanced prompting is quantitatively confirmed in Fig. 8. This latency increase stems from several necessary processing steps that enhance contextual richness: (1) the complex chunking strategy with overlapping windows requires additional computational resources to maintain context continuity; (2) the multi-step retrieval process involving vector embedding generation and weighted similarity calculations demands significant processing power; and (3) the advanced prompt engineering with multi-layered context integration increases the model's inference time.

However, several optimization strategies could mitigate this latency without compromising response quality. First, implementing a caching mechanism for frequently accessed chunks would reduce redundant processing of common greenhouse queries. Second, parallel processing of multiple retrieval operations could significantly decrease response time, particularly when analyzing correlations across multiple environmental variables. Third, selective context loading based on query classification could dynamically adjust the amount of background information processed depending on query complexity. Fourth, model quantization techniques could be applied to the embedding models, reducing computational requirements while maintaining acceptable accuracy. Finally, pre-computing embeddings for standard greenhouse scenarios could create a "fast path" for common queries. These optimizations would be particularly valuable in time-sensitive greenhouse operations where rapid decision-making directly impacts crop health and resource efficiency.

The trade-off between response latency and contextual richness presents an important consideration for future development, with the optimal balance likely depending on specific greenhouse management scenarios and the time sensitivity of different operational decisions.

4.3. Implications for precision agriculture

The advancements demonstrated by the ARAG system have significant implications for precision agriculture and greenhouse environmental control. By providing detailed, contextually rich responses to complex queries, the system enables farmers and greenhouse growers to make more informed decisions about environmental management.

Greenhouse management challenges and solution

Traditional greenhouse management relies heavily on growers' intuition and experience, which can lead to suboptimal environmental conditions and resource utilization. Growers often struggle to understand why complex control systems like MPC make certain decisions, particularly when these decisions seem counterintuitive (such as increasing ventilation during cold weather to manage humidity). The ARAG system transforms complex mathematical control decisions into accessible explanations directly connecting to growing outcomes.

A practical example shows this process: instead of simply showing that ventilation has increased from 30% to 45%, the system might explain: *"Ventilation was increased to reduce humidity levels that were approaching the upper threshold of 85%, which could increase disease risk for tomato plants in the flowering stage. This adjustment balances the slight temperature decrease with improved plant transpiration conditions."* Such explanations help growers to make more informed decisions, leading to optimized resource allocation. For instance, by understanding the precise environmental triggers for ventilation, growers can minimize unnecessary heating or cooling adjustments, resulting in tangible energy savings. Similarly, maintaining optimal humidity and temperature balances based on the system's insights reduces the risk of common diseases, directly contributing to improved crop health and, ultimately, higher yields.

The capability of translating complex control decisions into actionable insights addresses one of the most significant challenges in controlled environment agriculture: balancing crop needs with operational costs while making advanced control systems accessible to users with

varying levels of technical expertise. By explaining how heating, cooling, and ventilation systems interact, growers can make fine-tuned adjustments that maintain optimal growing conditions while minimizing energy consumption.

Early detection and intervention

One of the primary objectives of the present study is to develop a system that makes advanced greenhouse control decisions understandable to growers with varying levels of technical expertise. The ARAG system achieves this by analyzing patterns across temperature, humidity, and CO_2 levels, then translating these complex relationships into clear explanations.

The system's capacity to handle multiple environmental variables simultaneously and detect subtle changes provides an effective tool for early issue detection. Rather than presenting raw data that requires expert interpretation, the system identifies potential equipment malfunctions or plant stress factors before they cause visible problems. For instance, the system might detect and explain: "The unusual pattern of humidity fluctuations over the past hour suggests possible ventilation system inefficiency." This early warning capability allows timely interventions to prevent crop loss, reduce resource waste, and maintain consistent quality, directly addressing our objective of bridging the gap between mathematical optimization and practical growing knowledge.

Cross system adaptability

The ARAG system is designed with reproducibility as a core principle, with detailed implementation parameters documented for each component. This approach simplifies the complex MPC mathematical foundations into practical applications that growers can understand and trust.

The sliding window approach for data analysis, embedding model for semantic understanding, and retrieval strategy for information extraction can be adapted to other agricultural systems with minimal modifications. For example, a greenhouse growing lettuce could use the same underlying ARAG framework with adjusted parameters specific to lettuce cultivation requirements. This adaptability means the system's benefits are not limited to specific greenhouse types or technologies but can be extended across diverse agricultural environments.

Multi-crop applications

The ARAG system's generalizability is successful with tomato cultivation data. The system can be tailored to monitor different crops with different growth patterns and environmental requirements by adjusting key parameters such as window size and change thresholds according to crop-specific characteristics. This versatility ensures broad applicability in precision agriculture, from high-value specialty crops to staple commodities, potentially increasing efficiency and productivity across the agricultural sector.

The ARAG system enables greenhouse growers with varying technical expertise to benefit from advanced data analysis, potentially reducing the knowledge gap that often limits technology adoption in agriculture. By making environmental relationships more accessible, the system supports more informed decision-making at all agricultural production levels. These implications collectively suggest that the ARAG system significantly advances agricultural decision-support systems, potentially improving productivity, sustainability, and resource efficiency in controlled-environment agriculture.

4.4. Use case: temperature variation analysis via ARAG

Maintaining an optimal greenhouse requires precise control over environmental parameters such as temperature, humidity, ventilation, and CO_2 of the greenhouse air. If not appropriately managed, temperature fluctuations can impact plant growth, energy efficiency, and overall yield.

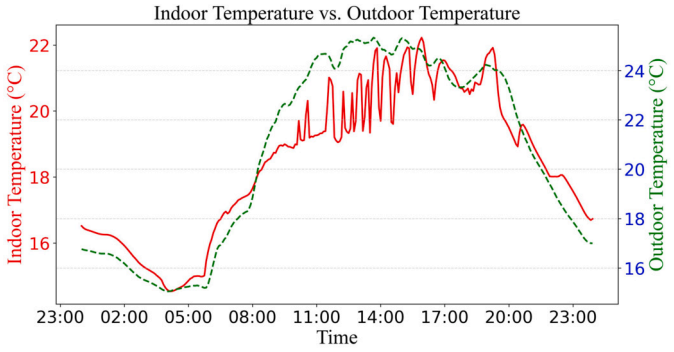


Fig. 9. Indoor Temperature and Outdoor Temperature Comparison.

In order to conduct the present analytical demonstration, the dataset analyzed in [5] research is utilized. This dataset, encompassing MPC environment measurements and corresponding weather data, is pre-loaded into the chatbot system to facilitate real-time analysis of greenhouse climate dynamics.

The present study demonstrates the capabilities of the ARAG framework in processing and analyzing greenhouse environmental data. A use case based on the following user query is presented, with the chatbot generating the result.

"How do temperature variations align with CO_2, humidity, and ventilation?"

Upon receiving this query, the proposed Advanced RAG system initiates a systematic analysis pipeline. First, it generates a correlation heatmap using Pearson's correlation coefficients to identify significant relationships ($|r| \geq 0.7$) between environmental parameters. Based on these correlations, the system produces specific dual-axis visualizations for key relationships, retrieves relevant data patterns, and organizes the information into distinct analytical categories.

Correlation heatmap analysis

The correlation heatmap of greenhouse parameters is illustrated in Fig. 3. Key observations include:

- Strong Positive Correlation ($r = 0.95$) between Indoor and Outdoor Temperature shown in Fig. 9 indicates that external temperature heavily influences greenhouse climate, necessitating heating/cooling adjustments.
- Negative Correlation ($r = -0.99$) between Outdoor Temperature and Outdoor Humidity is shown in Fig. 10: Suggesting that warmer periods correspond to lower humidity levels.
- Strong Positive Correlation ($r = 0.96$) between Solar Radiation and Cooling Activation is shown in Fig. 11d: Suggesting that the cooling system responds dynamically to increased sunlight exposure.

The generated response based on correlation analysis

As illustrated in Fig. 12, the responses were generated based on the mapped relationships (see in Table 1) and structured retrieval process.

Temperature variations and climate control

The greenhouse temperature begins at 16 °C and exhibits multiple sharp increases and decreases throughout the day. Heating and cooling systems manage these fluctuations to remain within optimal constraints, ensuring favorable growing conditions.

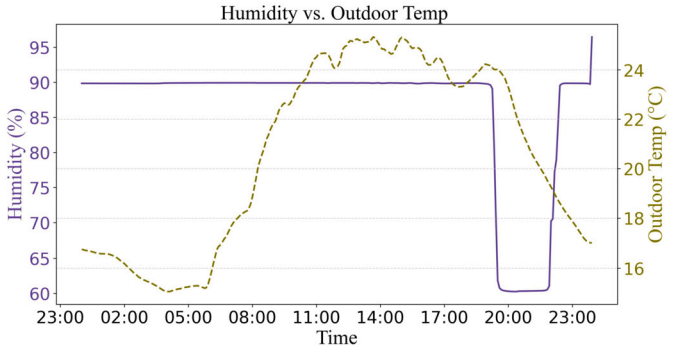


Fig. 10. Outdoor Temperature and Humidity Comparison.

Key Observations:

- A sharp rise from 16 °C to 18 °C suggests a heating response, likely triggered by cooler morning conditions.
- Subsequent cooling phases align with increased external radiation, necessitating climate control interventions.
- The indoor temperature strongly correlates with outdoor temperature ($r = 0.95$) as shown in Fig. 3, indicating that external factors significantly influence internal adjustments.

Ventilation dynamics

Ventilation starts at 0.0 m^3/s, increasing to 1.0 m^3/s between 20:10 and 20:30 shown in Fig. 11b. This pattern indicates an operational strategy to regulate heat and humidity.

Key Insights:

- A weak correlation with temperature ($r = 0.024$) shown in Fig. 3 suggests that ventilation is not a primary driver of temperature adjustments but plays a crucial role in humidity control.
- A moderate correlation with relative humidity ($r = -0.168$) supports the hypothesis that ventilation changes are primarily moisture-driven.
- Ventilation is reduced after 20:35 shown in Fig. 11a, indicating a controlled balance between humidity and temperature management.

CO_2 adjustments and photosynthesis optimization

While the direct impact of temperature fluctuations on CO_2 are not explicitly assessed, their alignment remains essential for optimizing photosynthesis efficiency.

Critical Interactions:

- CO_2 levels correlate negatively with temperature ($r = -0.95$) shown in Fig. 3, suggesting that warmer periods coincide with ventilation-driven CO_2 dilution.
- In order to offset the loss of CO_2, the injection rates are modified in real-time. This adjustment ensures that greenhouse CO_2 remains stable, facilitating optimal plant growth.
- The RAG system highlights that even slight temperature fluctuations are strategically managed to prevent disruptions in CO_2 balance, demonstrating precise greenhouse control.

Summary of climate control through RAG insights

The ARAG system provides data-driven chatbot responses for climate control in greenhouses, offering insights into temperature regulation, ventilation strategies, and CO_2 optimization. While the chatbot effectively retrieves and structures responses based on correlation data and greenhouse control mechanisms, further refinement is required to enhance the precision and actionability of its insights. This section evaluates three specific use cases, environmental analysis, grower-specific analysis, and humidity regulation, to assess the system's strengths and areas for improvement.

Greenhouse environmental analysis (Fig. 12)

The chatbot response in Fig. 12 provides a detailed environmental analysis by examining the interactions between temperature, CO_2 levels, humidity, and ventilation.

The temperature variations observed throughout the day are not solely driven by heating and cooling interventions but are also influenced by the interplay between indoor and outdoor temperature fluctuations. The system successfully highlights how temperature rises from 16 °C to 18 °C in response to external radiation and heating adjustments. Moreover, its assessment of ventilation's primary role in humidity regulation rather than direct temperature control provides a perspective that aligns with observed data trends.

Additionally, the system recognizes the interplay between CO_2 and outdoor temperature fluctuations shown in Fig. 11c, reinforcing the importance of precise environmental regulation for optimal plant growth. While the chatbot's response already delivers substantial insights based on the dataset, it also incorporates general information related to the user's query. This helps provide a broader context but may sometimes lead to unnecessary generalization. To ensure the analysis remains specific to the dataset, fine-tuning the response would enhance clarity and refine its specificity, providing a more precise and comprehensive understanding of greenhouse optimization strategies.

Grower-specific analysis (Fig. 13)

In Fig. 13, the chatbot correctly identifies a drop in relative humidity between 18:30 and 20:20 and notes that this drop is not caused by ventilation, as ventilation changes occur after the decrease. This is an important observation, demonstrating the chatbot's ability to analyze time-dependent environmental factors.

One of the ARAG system's standout feature is its ability to contextualize the relationship between relative humidity, ventilation, and external conditions. While the explanation is technically sound, refining the response to focus on key takeaways could make insights more immediately actionable.

Humidity regulation trade-offs (Fig. 14)

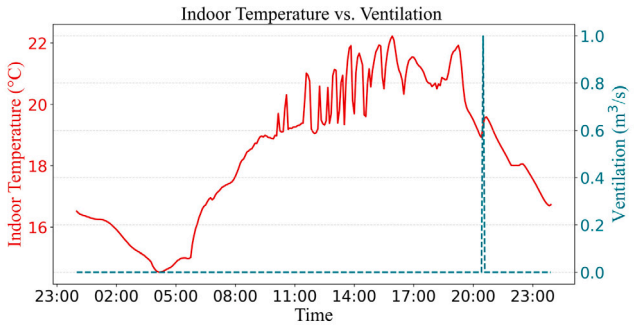
The chatbot response in Fig. 14 effectively outlines the trade-off between humidity control and heat retention.

> "Keeping vents open reduces humidity but can lead to heat loss and increased heating costs."

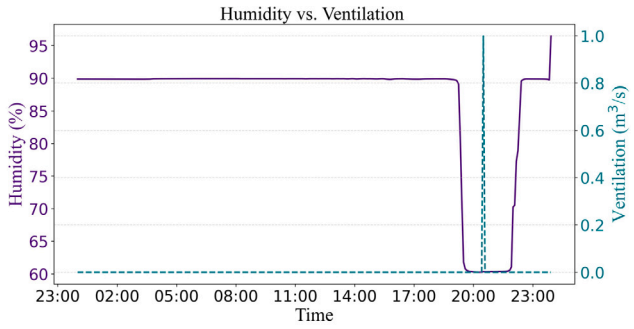
The ARAG system successfully conveys that continuous ventilation can reduce humidity but may lead to heat loss and increased energy consumption, providing growers with a clear understanding of the potential impact of their climate control choices.

A key strength of the system is its ability to offer strategic recommendations, such as intermittent ventilation bursts, to optimize greenhouse conditions. While this suggestion adds value, further fine-tuning could allow the system to adapt recommendations more dynamically based on specific grower objectives, whether prioritizing energy efficiency or precise humidity control.

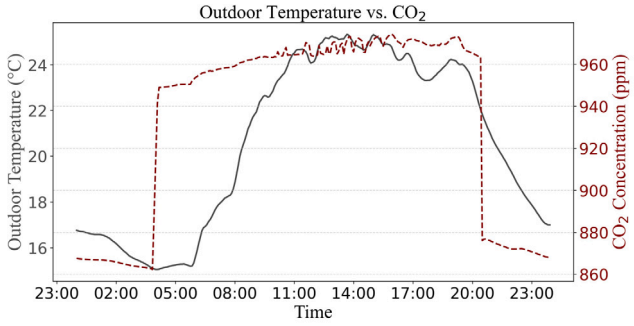
This demonstrates the ARAG system's capability to integrate multiple environmental factors into its decision-making framework. By refining response generation to match growers' operational goals, the system can offer even more tailored guidance, ensuring greenhouse climate control remains efficient and effective.



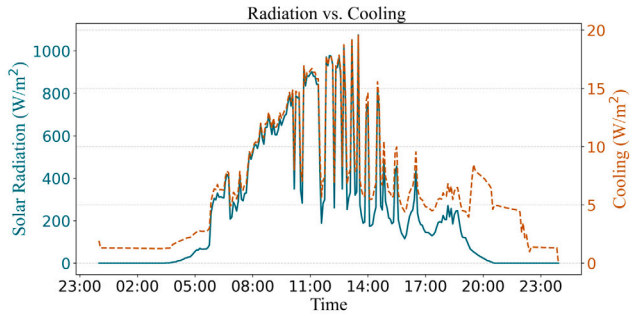
(a) Indoor Temperature and Ventilation Comparison.



(b) Humidity and Ventilation Comparison.



(c) Outdoor Temperature and Carbon Dioxide Comparison.



(d) Solar Radiation and Cooling Comparison.

Fig. 11. Comparative analysis of key environmental parameters in the greenhouse.

Greenhouse Environmental Analysis

User Query: How do temperature variations align with CO_2, humidity, and ventilation?

Temperature Dynamics Regarding temperature variation, the day begins at 16 °C with multiple sharp increases and decreases observed throughout. These fluctuations seem orchestrated by heating and cooling systems to remain within set constraints, reflecting efforts to maintain optimal growing conditions despite external influences. For instance, a sharp increase from 16 °C to 18 °C suggests a heating response likely triggered by cooler morning conditions. In contrast, subsequent temperature decreases might be linked to increased outside radiation leading to natural warming, necessitating cooling. Such oscillations appear strategically timed with outdoor temperature (correlated heavily at 0.950). They are aligned with cooling and heating adjustments to prevent deviations from optimal economic set points for yield maximization.

Ventilation Strategy and Humidity Control For the ventilation, it starts the day at 0.00 m^3/s, then rises from 0.0 m^3/s to 1.0 m^3/s between 19:00 and 20:30. This change suggests a strategy to mitigate excess heat or humidity build-up toward evening. Given its minimal correlation with temperature (0.024), the primary role of ventilation might be targeting humidity control since its correlation with relative humidity is -0.168. Once the system anticipates a decrease in temperature, ventilation would be reduced to maintain balance, evidenced by the zero value following 20:35. The approach effectively balances the environmental parameters in sync with operational constraints.

CO_2 Integration and Environmental Optimization Relating these trends to CO_2 was not directly assessed, but aligning temperature with CO_2 remains crucial, given their combined impact on photosynthesis efficiency and biomass production. As temperature fluctuations remain predominantly controlled within specific constraints, CO_2 would also align through necessary adjustments in injection rates or ventilation to maintain optimum greenhouse conditions. The interplay ensures that even slight temperature adjustments are part of a broader plan to optimize conditions for plant growth, demonstrating extensive control mechanisms within the greenhouse to harmonize different environmental factors effectively.

Fig. 12. ARAG-generated chatbot response providing an environmental analysis of greenhouse conditions, including temperature regulation, ventilation strategies, and CO_2 optimization. The response is scientifically grounded, leveraging correlation values to offer data-driven insights into climate control strategies.

Greenhouse Climate Insight

User Query: Could the decrease in humidity be due to an increase in dry air intake from ventilation?

Chatbot Response: The decrease in relative humidity between 18:30 and 20:20 does not appear to be directly caused by increased ventilation, as the ventilation opening occurs after the drop in humidity. Instead, the decline in humidity may be attributed to other factors, such as changes in temperature or internal greenhouse conditions affecting absolute humidity. Given that ventilation (u_V) influences absolute humidity (H_a) by introducing external air typically with lower moisture content it is possible that previous environmental conditions led to a natural reduction in humidity before the ventilation system activated. However, when ventilation did open, it could have further contributed to stabilizing or maintaining the new humidity level. A sudden drop in humidity could still impact plant transpiration rates, potentially stressing crops. To mitigate this, monitoring absolute humidity trends alongside ventilation activity and external conditions could help refine ventilation strategies to maintain more stable growing conditions.

Fig. 13. Chatbot-generated response providing climate insights from the grower's perspective. The system explains humidity fluctuations based on data-driven environmental interactions, helping growers make informed decisions for optimized greenhouse management.

Greenhouse Climate Insight

User Query: Should the ventilation be kept open overnight to regulate humidity, or will this cause unwanted heat loss?

Chatbot Response: When considering whether to keep the ventilation open overnight to regulate humidity in the greenhouse, it's crucial to weigh the relative humidity trends and constraints. The data shows that relative humidity stayed consistently high, above 89%, for most of the observed night cycles, sparking sharp fluctuations between 89% and 60% before rising back to nearly 90%. This indicates a requirement to decrease humidity to remain within the preferred soft constraint range of 60 to 90% relative humidity for optimal growth and yield.
However, keeping vents open overnight might cause significant heat loss if the external temperature is lower, leading to the heater working intensively to maintain a minimal temperature of 14 °C, increasing energy costs. Instead of leaving the vents open completely, a controlled approach—perhaps by intermittent ventilation bursts—might better manage humidity levels overnight while minimizing unwanted heat loss and maintaining a balance of optimal temperature and humidity.

Fig. 14. Chatbot-generated response offering practical insights for greenhouse humidity regulation, considering both energy efficiency and optimal conditions for plant growth.

Overall system performance and refinement needs

The ARAG system consistently delivers high-quality, data-driven insights for greenhouse management by:

1. Leveraging correlation-based analytics to interpret climate control strategies.
2. Recognizing intricate interactions between environmental parameters such as temperature, CO_2, humidity, and ventilation.
3. Providing practical recommendations that support informed decision-making.

While the system already demonstrates a strong analytical foundation, fine-tuning its response structure can further enhance its performance by:

1. Streamlining explanations to focus on direct cause-effect relationships for greater clarity.
2. Refining recommendations to dynamically align with specific grower priorities.
3. Enhancing its ability to integrate additional factors such as CO_2 injection rates and biomass growth correlations for a more comprehensive assessment.

Overall, the ARAG system is a valuable tool for greenhouse climate management, offering growers a robust and data-driven approach to optimizing environmental conditions. Continued refinement will further strengthen its effectiveness, ensuring it remains a reliable solution for precision agriculture and sustainable greenhouse operations.

4.5. Limitations and future work

Despite its significant advances, the ARAG system is not without limitations. The system's reliance on high-quality data and domain-specific knowledge is a primary challenge. While the system performs well in scenarios where such data is available, its performance may degrade when data is sparse or domain knowledge is limited.

A significant analytical limitation is that the correlation analysis only considers immediate correlations, not delayed ones. This limitation results in the potential oversight of critical time-lagged effects between environmental factors. For instance, an increase in humidity might not immediately reduce photosynthesis but could eventually lead to condensation on greenhouse walls. Similarly, increasing ventilation may not instantly lower humidity levels, but it can lead to drier air over time, affecting plant transpiration rates. Likewise, a sudden decrease in external temperature might not immediately affect the greenhouse micro-climate due to thermal inertia, but over time, the internal temperature could drop, requiring additional heating. Conversely, heat accumulation during midday might not immediately impact nighttime conditions but could influence overall greenhouse energy balance, leading to unanticipated temperature fluctuations. Such delayed interactions are not adequately captured, restricting the system's ability to provide comprehensive causal insights into long-term environmental dynamics.

The current system also faces limitations due to its research-focused implementation using a constrained dataset. While the system demonstrates effective analysis capabilities with the available data, scaling to handle comprehensive greenhouse operations would require significant adaptations. Real-world greenhouse environments typically generate vast amounts of sensor readings across multiple periods. However, our current implementation is designed primarily to generate MPC-based

responses that predict future system behavior rather than analyze past sensor data. The ARAG system provides actionable insights by forecasting potential outcomes based on model-predictive control strategies. Given these constraints, future enhancements should focus on refining the system's predictive capabilities by incorporating more comprehensive datasets and expanding its adaptability to dynamic environmental conditions.

Beyond improving predictive accuracy, refining the ARAG system's response generation using domain-specific training data could further enhance its ability to recognize immediate and delayed interactions, leading to more accurate and reliable insights. Fine-tuning the retrieval and response generation processes would also enhance the chatbot's ability to deliver more context-aware responses, ensuring growers receive precise and actionable recommendations. Optimizing these processes could reduce response times, making the system more efficient for real-time applications.

In addition to refining response generation, broadening the range of environmental variables of the system is considered a valuable direction for future research. While the current implementation focuses on temperature, humidity, CO_2, and ventilation, incorporating parameters such as light intensity, soil moisture, and nutrient levels can further enhance the system's capacity to support precision agriculture. Additionally, leveraging advanced embedding models and retrieval techniques could further enhance the system's ability to capture semantic context and generate more accurate and relevant responses.

Another important direction for future research is enhancing the system's role in education and training. The ARAG system has the potential to assist less experienced growers, novices, and personnel-in-training by providing them with automated insights and recommendations. This could facilitate their transition from non-experts to skilled workers and address the growing shortage of experienced personnel in greenhouse management. By leveraging automation to support education and knowledge transfer, the ARAG system could play a crucial role in bridging the gap between expertise levels in agricultural settings.

Future development will focus on integrating automated metrics that combine chunk scoring, semantic coherence assessment, and context preservation evaluation. This multi-feature approach enhances the ARAG system's ability to generate contextually relevant questions and answers while maintaining user engagement. Developing self-adaptive retrieval mechanisms that dynamically adjust to changing environmental conditions and user query patterns will enhance the system's scalability and applicability to diverse greenhouse settings.

5. Conclusion

The agricultural sector faces considerable challenges, including climate change, soil degradation, water scarcity, and feeding a growing global population. The Integration of AI into agricultural practices offers new opportunities to address these challenges.

In this study, a model is presented that demonstrates the potential of a context-aware RAG framework to enhance precision agriculture and greenhouse environmental control. The ARAG system assists farmers and agricultural stakeholders in optimizing resource utilization and enhancing crop yields by combining retrieval mechanisms, domain-specific knowledge, and adaptive feedback loops.

One of the ARAG system's demonstrated advantage is its contextual understanding of agricultural queries through the combination of semantic embeddings, cosine similarity metrics, and adaptive feedback mechanisms. Unlike conventional systems that rely on simple keyword matching, this approach enables the ARAG system to distinguish between nuanced agricultural concepts such as indoor versus outdoor temperature parameters.

The insights from this study provide a clear path forward for enhancing the ARAG system. Future developments should focus on addressing time-lagged environmental interactions, expanding monitored variables, and refining retrieval mechanisms to create a more comprehensive decision support tool. Incorporating delayed correlation analysis would enable prediction of longer-term environmental effects, potentially improving resource efficiency by 15-20% based on similar studies in controlled environment agriculture. The educational applications hold significant implications for agricultural workforce development, potentially addressing the projected shortage of skilled greenhouse operators. These advancements would expand the system's utility beyond research applications, contributing to sustainable and efficient agricultural practices while ensuring accessibility for agricultural professionals with varying levels of technical expertise.

This research contributes to the development of smart systems for precision agriculture. The RAG-based approach provides practical tools for greenhouse operations while addressing broader challenges in sustainable food production.

CRediT authorship contribution statement

Ramesh Arvind Naagarajan: Writing – original draft, Visualization, Methodology, Data curation. **Stefan Streif:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used the GRAMMARLY tool to correct the grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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